Nallabati Venkata Durga Sai Sarat Chandra

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EDUCATION

Amrita Vishwa Vidyapeetham — Amritapuri, India

B.Tech in Robotics and Artificial Intelligence Engineering — CGPA: 9.05 / 10.00 · Aug 2024 – May 2028 (Currently in 3rd Year)

Relevant Coursework: Multivariable Calculus, Linear Algebra, PDEs, Robotics Kinematics & Dynamics, Microcontrollers & RTOS, Algorithms, Machine Learning, Computer Vision, Digital Signal Processing.

TECHNICAL SKILLS

Languages & Autonomy: Modern C++ (C++17/20), Python (Expert), Embedded C, SQL, ROS 2 (Jazzy/Humble), CycloneDDS, OpenCV, FreeRTOS

Hardware & Silicon: STM32 (ARM Cortex-M), ESP32, Valetron A7670C 4G LTE, CAN 2.0B/CAN-FD, UART DMA, I2C, SPI, 240V Relays, KiCAD

Scientific ML & Edge AI: PINNs, Proper Orthogonal Decomposition (POD/SVD), SINDy, PyTorch, ANSYS Fluent CFD, AMD Ryzen AI NPU

PROFESSIONAL EXPERIENCE & SYSTEMS LEADERSHIP

CLUB MIRAI (Robotics & Embedded Systems Club) — Amrita Vishwa Vidyapeetham

Embedded Systems and Robotics Developer & Hardware Lead — Aug 2023 – Present

- Engineered modular ROS 2 node architecture with CycloneDDS zero-copy middleware for real-time 50 Hz closed-loop control.
- Implemented bare-metal STM32 firmware utilizing DMA-driven circular UART buffers at 921,600 baud with zero packet drop.

KEY ENGINEERING & RESEARCH PROJECTS (4 DISTINCT FLAGSHIP SYSTEMS)

1. Physics-Informed Neural Network (PINN) Non-Newtonian Hemodynamics Pipeline

PyTorch, SciPy, ANSYS Fluent CFD, Carreau-Yasuda Rheology, Adaptive NTK Balancing — Lead Developer · 2024 – Present

- Formulated Navier-Stokes forward/inverse physics-loss PINN inverting blood rheology parameters from pulsatile CFD velocity fields.
- Implemented Neural Tangent Kernel (NTK) trace balancing, achieving <2.4% relative L_2 error against transient ANSYS Fluent.

2. Reduced-Order Modeling (ROM) & Sparse Nonlinear Dynamics (SINDy)

Proper Orthogonal Decomposition (POD/SVD), STLSQ Sparse Regression, Python, NumPy — Lead Developer · 2024

- Constructed POD-Galerkin reduced basis capturing 99.2% turbulent kinetic energy across 12 empirical modes with 1200× speedup.
- Extracted sparse nonlinear ODEs via L_1-regularized STLSQ across 15,000+ spatial mesh points with zero structural divergence.

3. Evolution Edge — Autonomous Companion Mobile Robot

ROS 2 Jazzy, CycloneDDS, C++17, 5-DOF IK, OpenCV, LiDAR + 9-DOF IMU Sensor Fusion — Lead Architect · 2024 – Present

- Derived analytical & Damped Least-Squares (DLS) Inverse Kinematics for 5-DOF arm executing trajectory planning in $<50\,\mu$s.
- Integrated 2D LiDAR + 9-DOF IMU quaternion EKF state estimation achieving ± 1.8 cm positional tracking accuracy.

4. Valetron A7670C 4G LTE/GSM Smart High-Inductive Agricultural Telemetry

Embedded C++, STM32 ARM Cortex-M, FreeRTOS, Non-Blocking AT Parser, 240V Relays — Firmware Lead · 2024

- Programmed bare-metal/FreeRTOS firmware driving 4G LTE Cat-1 modem over DMA circular UART with <1.8s auto-failover.
- Hardened opto-isolated 240V relay switching with RC snubber transient suppression under 90V–480V inductive surges.

PEER-REVIEWED PUBLICATIONS & PREPRINTS (ALL PAPERS)

1. Development and Real-Time Implementation of a Vision-Guided Autonomous Electric Vehicle

IEEE International Conference on Robotics and Mechatronics (ICRM 2025) — Accepted & Presented · Jan 2025

- Designed lane-tracking perception running at 35+ FPS on ARM Cortex-A SBC with closed-loop PID lateral trajectory tracking.

2. 25-Day Coastal Field Performance of Solar Parabolic Dish Desalination System

Elsevier Desalination Journal Submission · Empirical Thermodynamics — First Author / Co-Investigator · 2024

- Logged 116°C focal receiver temperature, achieved 99.5% TDS reduction (3,850 to 18 mg/L), Adjusted $R^2 = 0.83$ ($p < 0.05$).

HONORS & COMPETITIVE AWARDS

- Global Finalist, IEEE IES GenAI Challenge (2024): Physics-guided diffusion models with strict thermodynamic loss.
- ISRO YUVIKA Space Science Fellow (2020): Top 0.1% nationwide selection by Indian Space Research Organisation.
- National Anveshika Experimental Physics (NAEST 2021): Qualified 2 national screening tiers under Prof. H.C. Verma.
- State Rank 497 / 150,000+, AP POLYCET (2022): Top 0.3% percentile in competitive engineering entrance.